Computer Graphics Engineer | Technical Artist | XR Developer

Turin, Italy •+39 351 307 8191 •colombo.federico12@gmail.com •linkedin.com/in/federico-colombo10
•federicocolombo12.github.io/PortFolio

Computer Graphics Engineer and Technical Artist specialized in real-time rendering, spatial computing (XR), and procedural tool development. Graduated with honors in Cinema and Media Engineering. Deep expertise in writing custom C++ rendering engines, optimizing GPU architectures (HLSL/GLSL), and leading multidisciplinary teams. Actively seeking to leverage hardcore engineering and project management skills to build scalable software architectures and high-fidelity interactive productions.

M.Sc. Cinema and Media Engineering *2024 – 2026*

Politecnico di Torino • Turin, Italy

- 110/110 Cum Laude (Graduating Dec 2026)

B.Sc. Cinema and Media Engineering *2021 – 2024*

Politecnico di Torino • Turin, Italy

- 100/110

Software Developer & AI Solutions Consultant *06/2026 – Present*

Intellecto IP • Turin, Italy / Remote

- Engineered a centralized personal banking aggregator in React, uniting multi-bank account feeds and financial data into a single, high-security reactive dashboard with modular state management.
- Provided strategic advisory on enterprise AI adoption; deployed Anthropic Claude LLM workflows into business processes to automate document analysis and operational decision-making.
- Automated repetitive internal workflows and project management pipelines across the Notion ecosystem via custom API integrations and scripts, eliminating manual overhead.

Technical Artist & Producer (XR Master Thesis R&D) *03/2026 – Present*

Lemons in the Room & Politecnico di Torino • Florence / Remote

- Architected a real-time bio-adaptive volumetric raymarching framework in HLSL for Meta Quest 3, rendering dynamic Signed Distance Fields (SDFs) driven by BLE biometric telemetry ( ).
- Engineered low-level GPU optimizations (Dynamic Bounding Volumes, adaptive ray stepping) locking (61.3% GPU time reduction, 2.91ms GPU time, 0B GC alloc).
- Authored shading on procedural volumes to eliminate triplanar UV seams, plus a sub-millisecond GPU Voxelizer (<0.8ms) for particle swarms.
- Architected a (`DAWMixerWindow`) inside Unity with live OLED oscilloscopes and auto-packet sniffing for multi-sensor streams (BLE, OSC, WebSockets, Serial); mapped autonomic biofeedback along the and modularized to official Unity Package (UPM) standards.

Technical Artist, Gameplay Programmer & Project Manager *11/2024 – 07/2026*

LEVEL UP LAB (PoliTo Student Team) • Turin, Italy

- Developed custom shaders in HLSL (retro CRT curvature, procedural water vertex displacement, cutaway systems) and core C# gameplay systems across team productions including *Line Operator* and *Spicy Mercante*.
- Modeled 3D assets in Blender and engineered full architectural refactorings (3D-to-2D engine transition, buoyancy physics, inventory systems).
- Coordinated multidisciplinary student pods of 6–7 developers as technical producer and project manager, enforcing Git LFS versioning and milestone sprints.

Teaching Assistant (GUI & Frontend Development) *10/2024 – 01/2025*

Politecnico di Torino • Turin, Italy

- Tutored students in UI/UX principles and frontend programming during weekly laboratory sessions.
- Wrote technical documentation and created high-quality instructional content to improve comprehension.

Video Artist & Post-Production Specialist *09/2023 – 06/2024*

RKH SRL • Turin, Italy

- Operated professional cameras (RED, Blackmagic 4K/6K) and assisted DOP with complex lighting setups.
- Handled editing and post-production workflows using DaVinci Resolve for corporate content.

- C++, C#, Python, HLSL/GLSL, JavaScript/TypeScript
- Unity, Unreal Engine 5, Custom OpenGL Frameworks, React
- Anthropic Claude (LLM Integration), Notion API, Workflow Automation
- Blender, DaVinci Resolve, Jira, Git
- Technical Direction, Project Management, Agile Milestone Sprints

---

- Engineered a Python add-on for Blender to procedurally generate parametric hanging cables across multi-object modular assets with scale-independent catenary physics and automated mesh conversion for game engines.
- Engineered a custom OpenGL 3D rendering framework from scratch. Implemented Deferred Shading, Physically Based Rendering (PBR), and Silhouette Parallax Occlusion Mapping (SPOM) to study low-level GPU pipelines.
- Implemented a real-time painterly NPR pipeline in Blender 5.3 Beta and EEVEE-Next via OpenUSD PointClouds, Geometry Nodes vector flow fields, and Kuwahara filtering.
- Developed procedural environments using PCG. Scripted a homing projectile system in Niagara and integrated MetaHumans with custom groom collision physics.
- Developed an immersive VR educational experience (Unity) and a 2D Metroidvania prototype emphasizing complex transformation mechanics. Managed end-to-end technical production.